

Project Design Phase-II

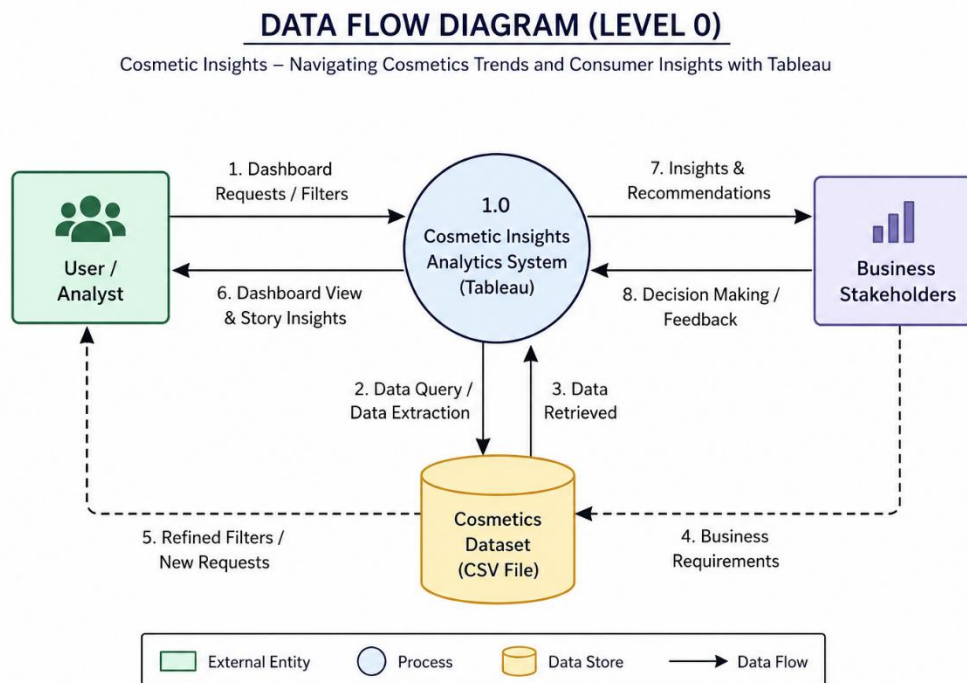
Data Flow Diagram & User Stories

Date	30 June 2026
Team ID	
Project Name	Cosmetic Insights: A Tableau-Based Analysis of Skincare Trends and Consumer Preferences
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD depicts the right amount of system requirement graphically, showing how data enters and leaves the system, what changes the information, and where data is stored.

DFD Level 0 — Cosmetic Insights



USER (Tableau Analyst) -> [1. Upload cosmetics.csv] -> Raw Data Store (cosmetics.csv, 1,472 rows)

Raw Data Store -> [2. Clean & Standardize in Excel] -> Cleaned Data Store

Cleaned Data Store -> [3. Connect to Tableau] -> Tableau Data Source ('cosmetics')

Tableau Data Source -> [4. Build Worksheets] -> 18 Worksheets (Brand Count, Category Distribution, Skin-Type Suitability, Price vs Rank, etc.)

18 Worksheets -> [5. Assemble Story] -> Story 1 (Storyboard)

Story 1 -> [6. Publish] -> Tableau Public Link / Dashboard Screenshots -> END USER (Buyer / Brand Analyst)

Data Fields Used

Field	Type	Description
Label	Text	Product category (Moisturizer, Cleanser, Face Mask, Treatment, Eye cream, Sun protect)
Brand	Text	Brand/manufacturer name (116 unique brands)
Name	Text	Product name
Price	Integer	Product price in USD (\$3 – \$370)
Rank	Decimal	Customer rating, 0–5
Ingredients	Text	Full ingredient list
Combination / Dry / Normal / Oily / Sensitive	Boolean (0/1)	Whether the product is suitable for that skin type

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Skincare Buyer	Category Browsing	USN-1	As a buyer, I can view product distribution by category so I know which categories have the most options.	Category distribution chart loads with correct counts	High	Sprint-1
		USN-2	As a buyer, I can filter products by skin type so I see only items	Filtering by Dry/Oily/Sensitive/Combination/Normal returns correct subsets	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
			suitable for my skin.			
	Brand Comparison	USN-3	As a buyer, I can view the top 10 brands by product count so I know which brands offer the widest range.	Top 10 brand chart sorts correctly by count	Medium	Sprint-1
	Price Analysis	USN-4	As a buyer, I can compare price against customer rank so I can judge value for money.	Scatter plot displays price vs. rank for all valid rows	High	Sprint-2
Brand / Retail Analyst	Brand Benchmarking	USN-5	As an analyst, I can view brand-wise average prices so I can benchmark my brand's pricing.	Average price per brand chart loads correctly	High	Sprint-1
	Category Revenue	USN-6	As an analyst, I can view category-wise total revenue (price proxy) so I can see	Category revenue totals match manual calculation	Medium	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
			which category drives the most value.			
	All-Skin-Type Brands	USN-7	As an analyst, I can view the top 10 brands suitable for all skin types so I can identify broadly compatible competitors.	Top 10 list correctly filters brands with all 5 skin-type flags set	Low	Sprint-2
Team / Presenter	Storyboard	USN-8	As a presenter, I can navigate a guided story so I can demo the findings in a logical order.	Story 1 loads all story points without error	High	Sprint-2